Accelerated
Intelligent
Systems Lab.

L2W7: Project 1

2023. 4. 17. Mon.

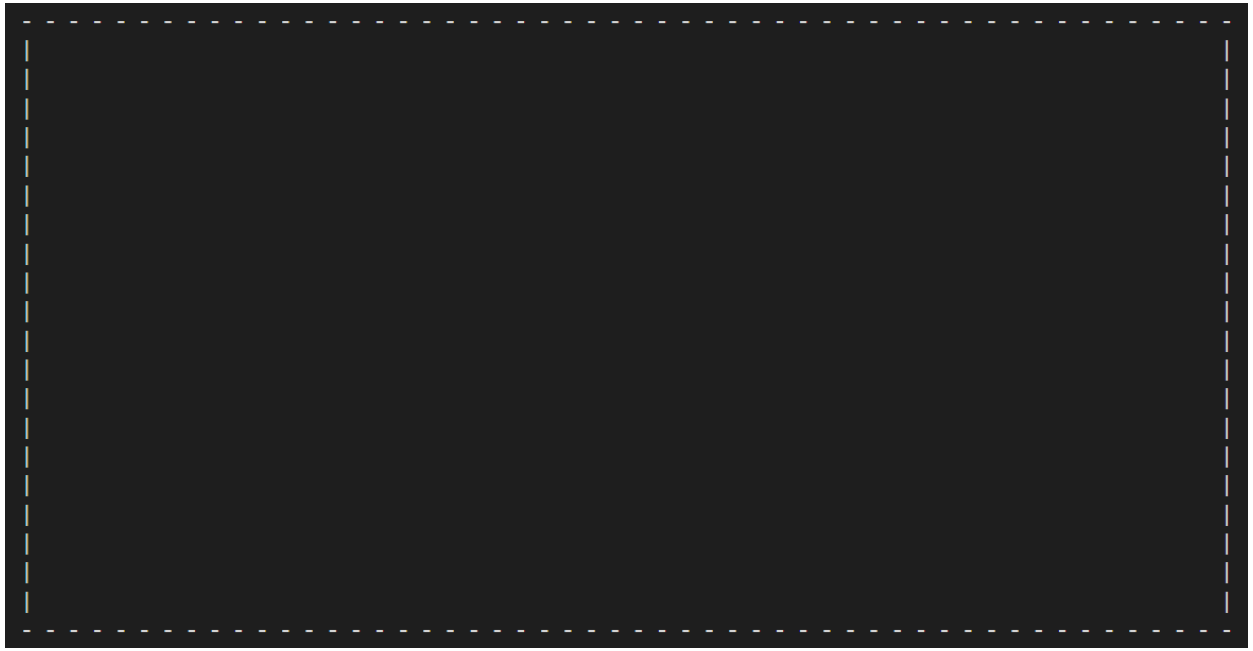
Contents

1. Assignment review
2. Project 1: Bulletin Board
   - Input, Output
   - Classes
   - Jobs (functions)
3. Attendance

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- In Project 1, your program has to maintain a bulletin board with multiple pages on it.
  - Classes
    - Page
    - Board
  - Jobs
    - Insertion
    - Deletion
    - Modification of page content
    - Modification of page position



Said “bulletin board”

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- Page
  - Defined in page.h
  - Class for pages to go on our board
    - x, y: position of the page on the board
    - width, height: width, height of page
    - id:
      - a unique id for each page
      - int between 0~32767
    - content:
      - char to be added to the board array
      - a single char a~z, A~Z
      - not unique

```
class Page {
    public:

    private:
        int x, y; // position of the page on the board
        int width, height; // width and height of the page
        int id; // unique id for each page
        char content;
};
```

There are no functions in the default page
Define them as you wish!

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- Board
  - Defined in board.h
  - Class for our bulletin board
    - num_jobs: total number of jobs
    - width, height: width, height of board
    - output: ofstream to output file
    - board:
      - A char array with the size of width*height
      - Think of it as a version of the board to be printed to the output file

```
class Board {
    public:
        Board(int num_jobs, int width, int height, ofstream& output_stream);
        ~Board();

        void print_board();
        void print_job(int job_idx, char job_type, int id);

        //job functions
        void insert_page(int x, int y, int width, int height, int id, int content);
        void delete_page(int id);
        void modify_content(int id, char content);
        void modify_position(int id, int x, int y);

    private:
        int num_jobs, width, height;
        ofstream& output;
        char* board;
};
```

2. Project 1: Bulletin Board

- Board
  - You should define…
    - insert_page()
    - delete_page()
    - modify_content()
    - modify_position()
    - And any other function you need!
  - Other functions are already defined
    - print_board()
      - Prints the contents of board to the output file
      - Needs to be called every time a modification to the board is made
    - print_job()
      - Prints the job index, the job type, and target page index

```
class Board {
    public:
        Board(int num_jobs, int width, int height, ofstream& output_stream);
        ~Board();

        void print_board();
        void print_job(int job_idx, char job_type, int id);

        //job functions
        void insert_page(int x, int y, int width, int height, int id, int content);
        void delete_page(int id);
        void modify_content(int id, char content);
        void modify_position(int id, int x, int y);

    private:
        int num_jobs, width, height;
        ofstream& output;
        char* board;
};
```

```
>> (1) Insert 8
- - - -
```

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- Input Arguments

```
./main input/input_0.txt output/output_0.txt
```

  - First argument: input file
  - Second argument: output file (for the program's result)
  - Do not modify this part
    - We will compile(g++ -o main main.cpp) & run your code with the following format (./main input.txt output.txt), and grade with the outputted results (output.txt)

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- Input File Format
  - Line 1: Board width
  - Line 2: Board height
    - The entirety of all pages will always be inside widthxheight
  - Line 3: Number of jobs
    - Will always be the same as the number of following lines
  - Following lines
    - Each line represents one job for one page (insert, delete, modify)
    - i: insert
    - d: delete
    - m 1: modify page content
    - m 0: modify page position

```
50
20
7
i 8 0 1 10 15 c
i 1 5 2 5 18 k
i 2 2 7 3 2 x
i 32 6 3 10 10 l
m 1 32 M
m 0 2 1 2
d 8
```

2. Project 1: Bulletin Board

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- Insert
  - Always starts with ‘i’
  - Info. on the newly inserted page is given in the following order
    - Page id (8)
    - Page position x (0)
    - Page position y (1)
    - Page width (10)
    - Page height (15)
    - Page content (c)

```
50
20
7
i 8 0 1 10 15 c
i 1 5 2 5 18 k
i 2 2 7 3 2 x
i 32 6 3 10 10 l
```

(0, 0)

x increases

y increases

```
>> (1) Insert 8
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
```

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- Insert

```
50
20
7
i 8 0 1 10 15 c
i 1 5 2 5 18 k
i 2 2 7 3 2 x
i 32 6 3 10 10 l
m 1 32 M
m 0 2 1 2
d 8
```

```
>> (2) Insert 1
- - - - - - - - - - - - - - -
|
| c c c c c c c c c c
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
- - - - - - - - - - - - - - -
```

```
>> (3) Insert 2
- - - - - - - - - - - - - - -
|
| c c c c c c c c c c
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c x x x k k k k k
| c c x x x k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
- - - - - - - - - - - - - - -
```

```
>> (4) Insert 32
- - - - - - - - - - - - - - - - - - - -
|
| c c c c c c c c c c
| c c c c c k k k k k
| c c c c c k l l l l l l l l l l
| c c c c c k l l l l l l l l l l
| c c c c c k l l l l l l l l l l
| c c c c c k l l l l l l l l l l
| c c x x x k l l l l l l l l l l
| c c x x x k l l l l l l l l l l
| c c c c c k l l l l l l l l l l
| c c c c c k l l l l l l l l l l
| c c c c c k l l l l l l l l l l
| c c c c c k l l l l l l l l l l
| c c c c c k l l l l l l l l l l
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
- - - - - - - - - - - - - - - - - - - -
```

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- Modify content
  - Always starts with ‘m 1’
  - Info. on the to-be-modified page is given the following order
    - Page id (32): the id of the page to be modified
      - An page id that is already on the board will always be given
    - Page content (M): change the content to the given char

```
50
20
7
i 8 0 1 10 15 c
i 1 5 2 5 18 k
i 2 2 7 3 2 x
i 32 6 3 10 10 l
m 1 32 M
m 0 2 1 2
d 8
```

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- Modify content
  - The board gets printed twice
  - Think of it as
    - 1) removing the target page
    - 2) putting the modified page back on the board
  - if page 32 was under another page,
    - 1) we have to remove the pages until there is no page above page 32
    - 2) remove page 32
    - 3) put the modified page back on
    - 4) put the pages removed from 1) back on in reversed order

    (more on this later)

```
>> (5) Modify 32
- - - - - - - - - - - - - - - - - - - -
|
| c c c c c c c c c c
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c x x x k k k k k
| c c x x x k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
- - - - - - - - - - - - - - - - - - - -
```

```
- - - - - - - - - - - - - - - - - - - -
|
| c c c c c c c c c c
| c c c c c k k k k k
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c x x x k M M M M M M M M M M
| c c x x x k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
- - - - - - - - - - - - - - - - - - - -
```

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- Modify position
  - Always starts with ‘m 0’
  - Info. on the to-be-modified page is given the following order
    - Page id (2): the id of the page to be modified
      - An page id that is already on the board will always be given
    - Page position x, y (1, 2): the target position for the page to move

```
50
20
7
i 8 0 1 10 15 c
i 1 5 2 5 18 k
i 2 2 7 3 2 x
i 32 6 3 10 10 l
m 1 32 M
m 0 2 1 2
d 8
```

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- Modify position
  - The board gets printed twice
  - Think of it as
    - 1) removing the page
    - 2) putting the same page back on the board at the target position
  - if page 2 was under another page,
    - 1) we have to remove the pages until there is no page above page 2
    - 2) remove page 2
    - 3) putting the same page back on the board at the target position
    - 4) put the pages removed from 1) back on in reversed order

  (more on this later)

```
>> (6) Modify 2
- - - - - - - - - - - - - - - - - - - -
|
| c c c c c c c c c c
| c c c c c k k k k k
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
- - - - - - - - - - - - - - - - - - - -
```

```
- - - - - - - - - - - - - - - - - - - -
|
| c c c c c c c c c c
| c x x x c k k k k k
| c x x x c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k M M M M M M M M M M
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
- - - - - - - - - - - - - - - - - - - -
```

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

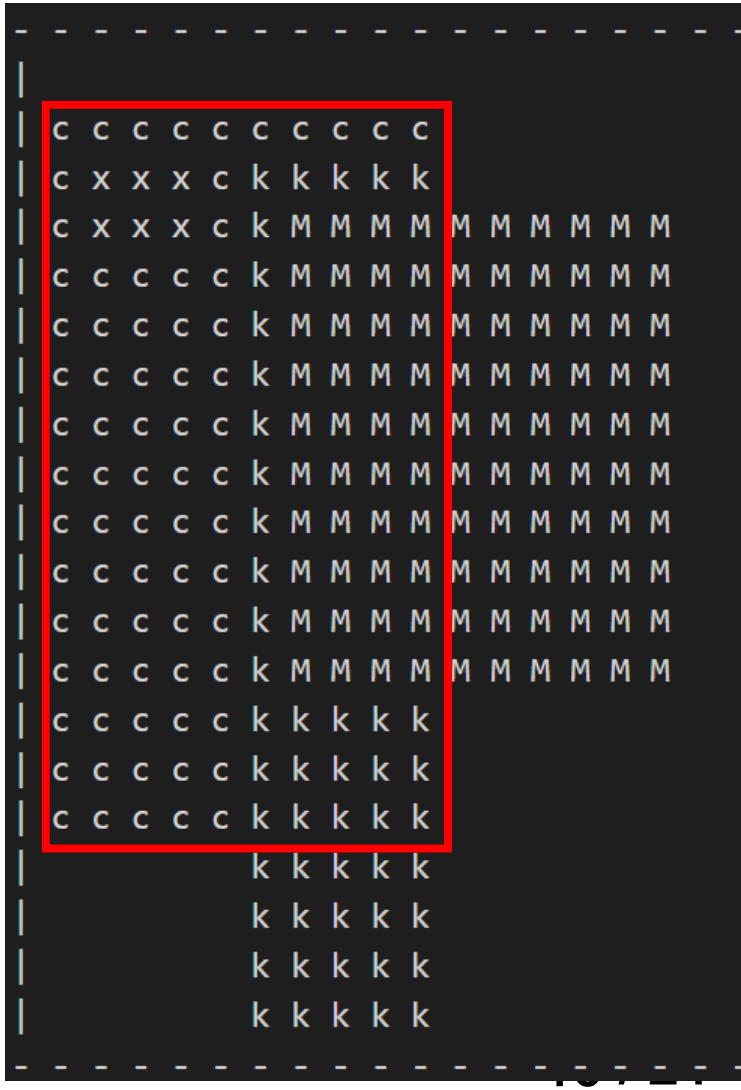
- Delete
  - Always starts with ‘d’
  - Info. on the to-be-deleted page is given the following order
    - Page id (8): the id of the page to be deleted
      - An page id that is already on the board will always be given

```
50
20
7
i 8 0 1 10 15 c
i 1 5 2 5 18 k
i 2 2 7 3 2 x
i 32 6 3 10 10 l
m 1 32 M
m 0 2 1 2
d 8
```

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

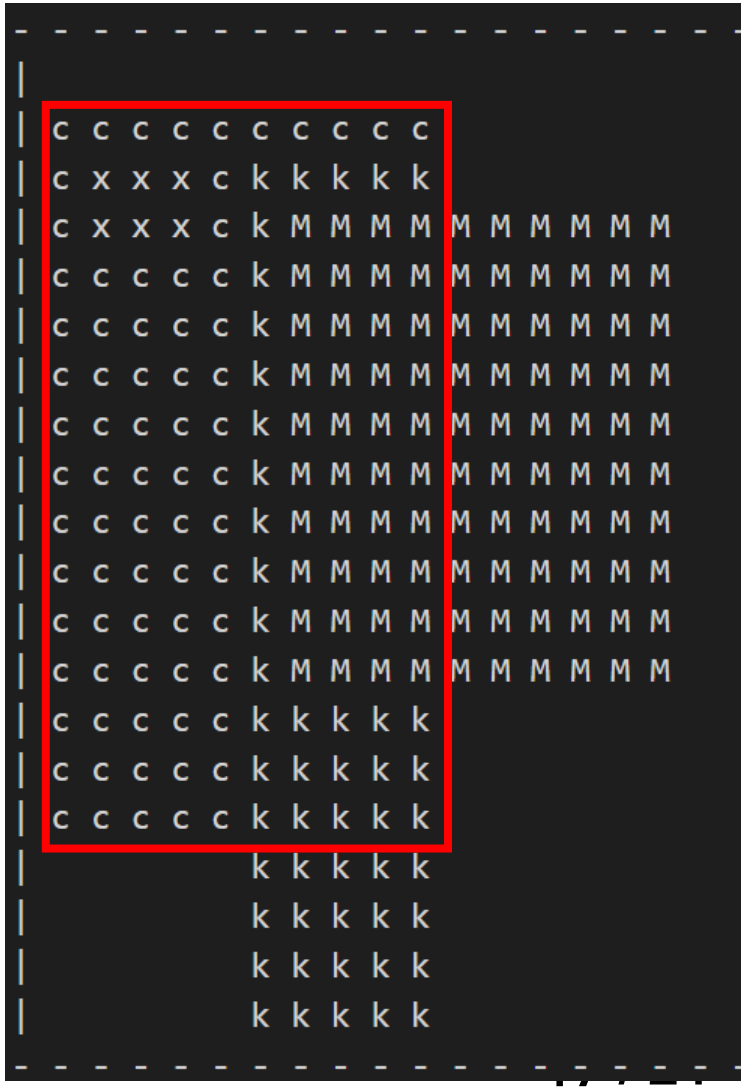
- Delete
  - Always starts with ‘d’
  - Info. on the to-be-deleted page is given the following order
    - Page id (8): the id of the page to be deleted
      - An page id that is already on the board will always be given
  - Uh-oh! Page 8 is below multiple pages!



2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

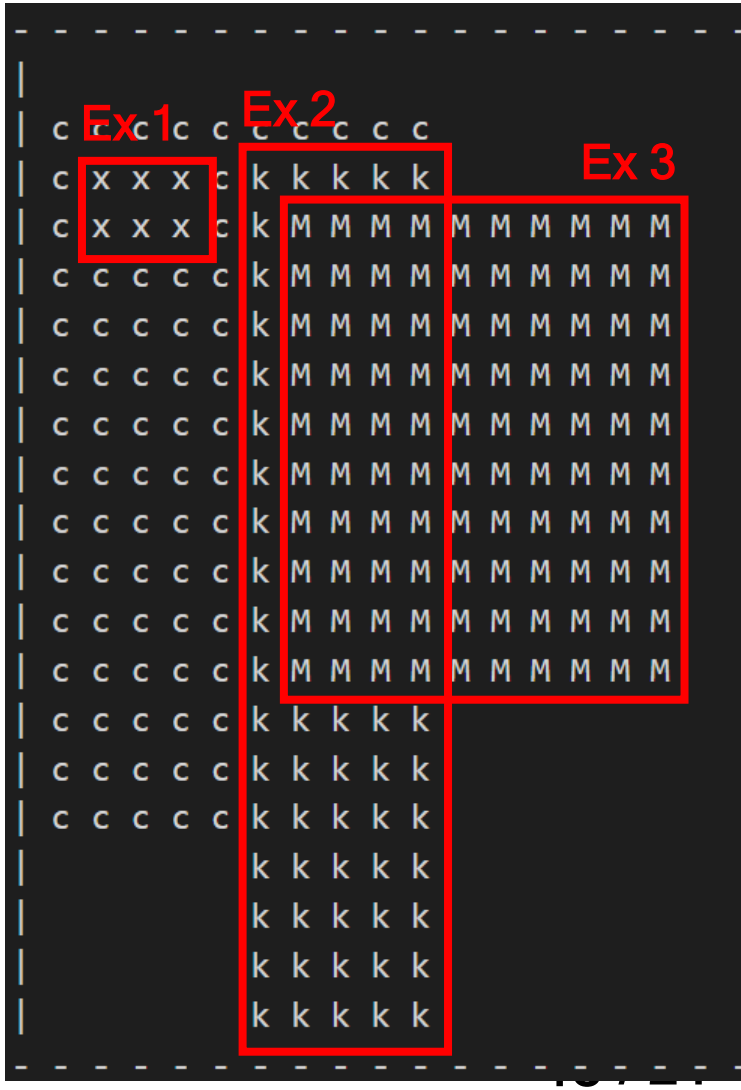
- When a target page is below (multiple) pages
  - 1) remove all pages above the target page
    - If there are multiple pages above, remove the page with the smallest page id first
  - 2) modify/delete target page
  - 3) re-post all pages back on the board in the reverse order of 1)
- This rule applies to <u>delete, modify content, modify position</u>



2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- When a target page is below (multiple) pages
  - 1) remove all pages above the target pagein specific order
    - What is the definition of “above”?
      - Page a is above page b only if page b is **directly** on top of page a for **at least one point**
        - **Ex 1.** Page x is above page c because it is **directly** on top of page c for 6 points
        - **Ex 2.** page k is above page c because although 20 points are not directly on top of page c, the **rest (at least one point)** are directly on top of page c
        - **Ex 3**. page M is **not above** page c because it is **not directly on top** of page c (page k is in between)



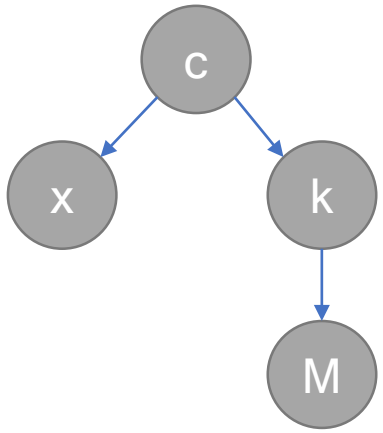
(on this section, for simplicity, the pages are named with their content)

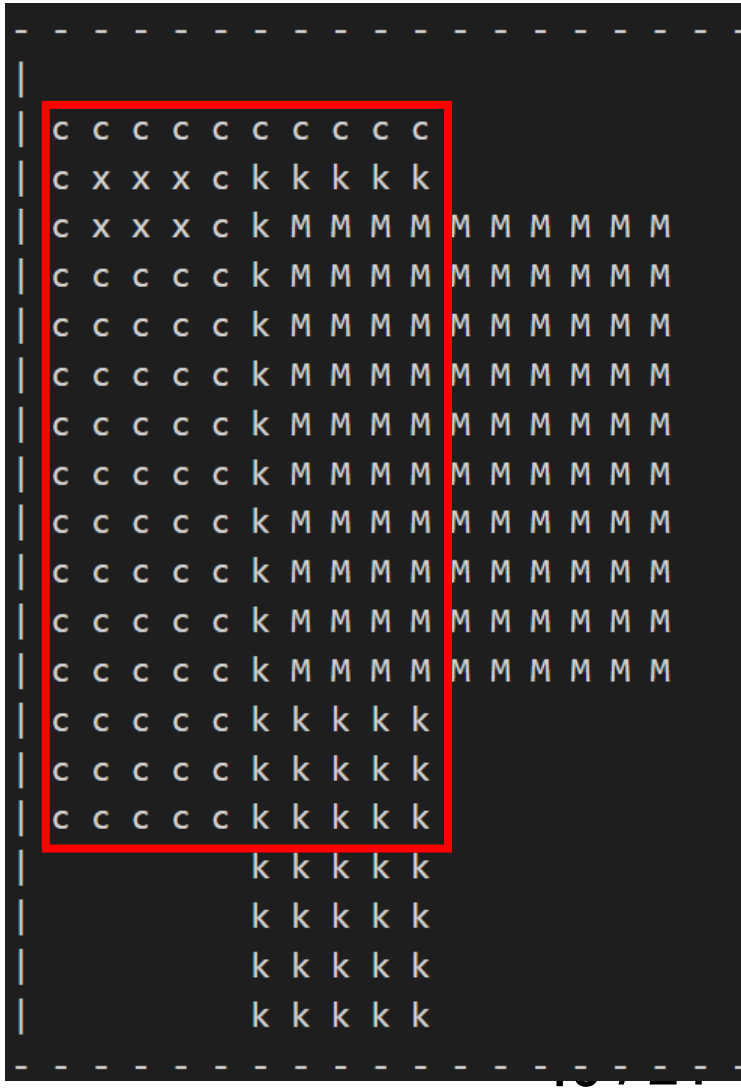
2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- When a target page is below (multiple) pages
  - 1) remove all pages above the target page in specific order
    - We can draw a graph with pages as nodes, and above-ness as edges
      - According to rule 1,
      - To remove c, we need to remove x & k
        - What do we remove first?
          - Page x's id: 2 > Page k's id: 1
          - we remove page k first!
            - But page k has page M on top!
              - Remove page M first!

Looks familiar? Recursion!

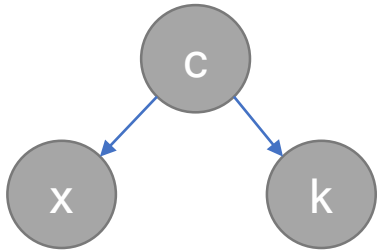




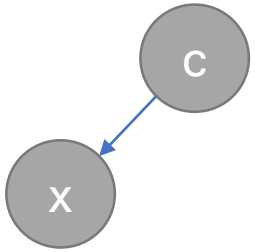
2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- Step-by-step Delete




```
>> (7) Delete 8
- - - - - - - - - - - - - - - -
|
| c c c c c c c c c c
| c x x x c k k k k k
| c x x x c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
| c c c c c k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
|           k k k k k
- - - - - - - - - - - - - - - -
```



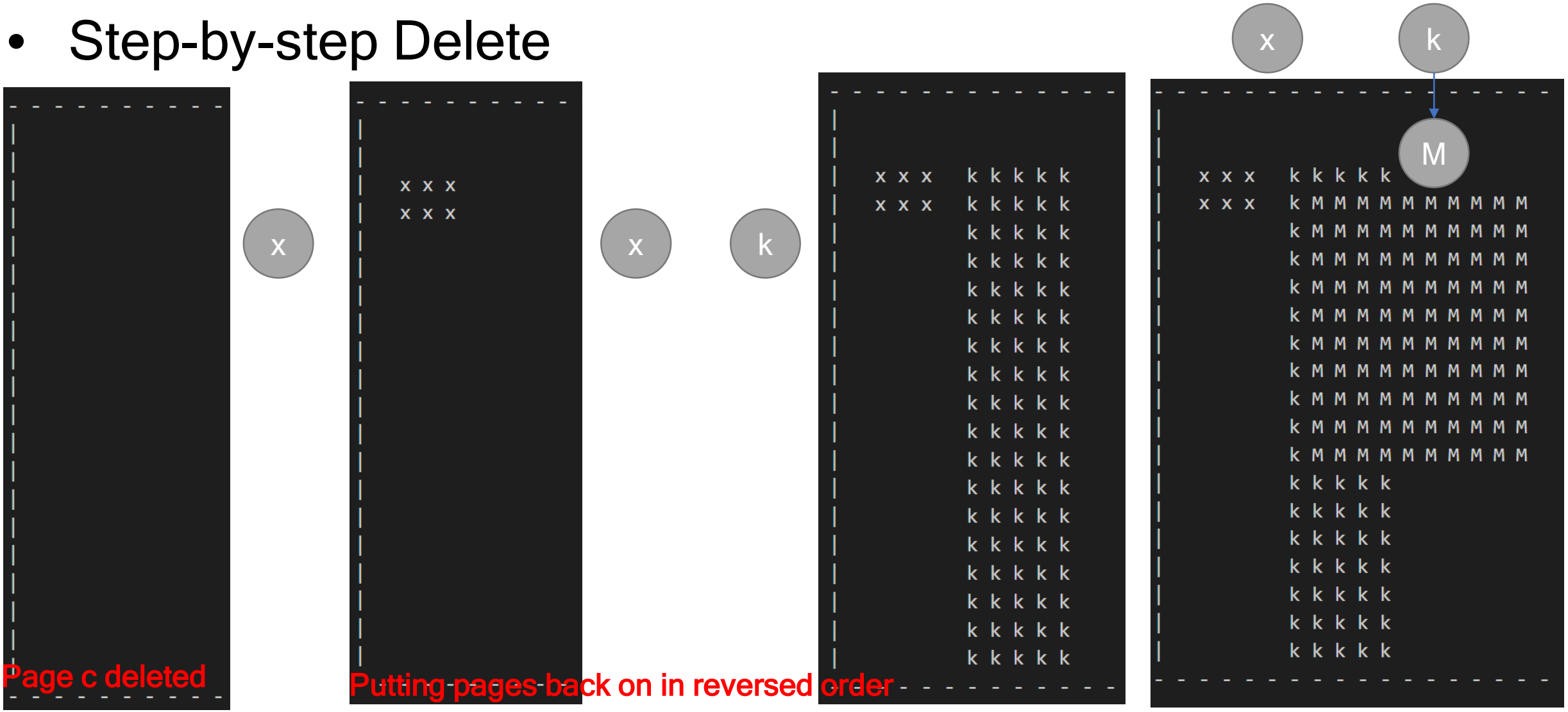

```
- - - - - - - - - - - - - - - -
|
| c c c c c c c c c c
| c x x x c c c c c c
| c x x x c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
|
|
|
|
- - - - - - - - - - - - - - - -
```

```
- - - - - - - - - - - - - - - -
|
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
| c c c c c c c c c c
|
|
|
|
- - - - - - - - - - - - - - - -
```

2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

- Step-by-step Delete



2. Project 1: Bulletin Board

Accelerated Intelligent Systems Lab.

Submit until eTL Project 1, by **17:50, 05.08 (Mon)**

- Late submission is allowed by 5.15 Monday 16:00 with 20% point deduction.
- Upload a **studentid_Project1.zip** file that contains
  - Project1 //directory with your source code
    - Project1 should contain a **.git directory with ≥ 10 meaningful commits**

3. Attendance

Accelerated Intelligent Systems Lab.

Submit Project1 code on replit

- You only have to press submit
  - No test cases
  - No need to check with TAs